Graphical Abstract: The Epistemic Fluency Trap in AI-Augmented Security Operations

Dual-Cue Epistemic Defeasibility Model (DCEDM) · Prompt-Isolated Telemetry Scorer (PITSA) · Two-Stage DECS ($N = 2{,}238{,}434$ Real Logs)

1. Socio-Technical Vulnerability

Indirect Telemetry Prompt Injection (ITPI):

- Composite alert $\mathcal{A}(e) = (\mathcal{A}_{\text{struct}}, \mathcal{A}_{\text{text}})$.
- Attacker injects benign clauses ($m \in \{1, 2, 4, 8\}$) into $\mathcal{A}_{\text{text}}$ while $\mathcal{A}_{\text{struct}}$ logs active intrusion.

Unpartitioned Recall Collapse (Prop. 1):

- **OTRF ($N_{\text{test}} = 13{,}117$):** LogReg $85.8\% \to 25.2\%$; Dense MLP $65.8\% \to 0.0\%$; NB/RF/SubLM $\to 0.0\%$.
- **SALAD-SOC ($N_{\text{test}} = 125{,}035$):** Clean 77.5% to 84.9% drops to 2.9% (LogReg), 4.0% (SubLM), 27.5% (MLP).

Epistemic Fluency Trap (Theorem 1):

- High narrative fluency ($\Delta F = 0.70$) shifts posterior log-odds by $-\kappa_F \lambda_F \Delta F$, inflating CMR($\Pi_2$) by $+14.97$ pp (OTRF) and $+26.02$ pp (SALAD).

2. PITSA & Two-Stage DECS (DP1 to DP3)

DP1: Sub-Channel Air-Gap (Π_{struct}):

- Discards $\mathcal{A}_{\text{text}}$ entirely ($\nabla_{\mathcal{A}_{\text{text}}} L(e) \equiv 0$).
- Scores 5 structural pillars in 2.12 to 4.31 μs/event.

DP2: White-Box Toulmin Defeater ($k^*(e)$):

- Additive LLR $L(e) = \sum_{k=1}^{5} \ell_k(a_k(e))$.
- Stage 1 Anchor $\mathcal{H}_1(e)$ before narrative; Stage 2 Rebuttal $\mathcal{R}_2(e)$ highlights $k^*(e) = \arg\max_k \ell_k(a_k(e))$.

DP3: Endogenous Habituation Gate (τ^*_{LLR}):

- Fires only when $\hat{y}_{\text{LLM}} = 0 \wedge L(e) > \tau_{\text{LLR}}$.
- Cuts warning banner volume by 95.92% (OTRF) and 79.87% (SALAD-SOC), preserving $\gamma_{\text{eff}}(\tau)$.

3. Multi-Corpus Validation ($N = 2.24$M)

100% ITPI Invariance Across 3 Corpora:

- **OTRF:** AUROC $= 0.9807$, Rec $= 81.94\%$ (0.00 pp loss; McNemar $\chi^2 \geq 86.01, p < 10^{-4}$).
- **SALAD-SOC:** AUROC $= 0.9380$, Rec $= 68.68\%$ (85.74% on `hard`; $\chi^2 \geq 8{,}361, p < 10^{-4}$).
- **LMD-2023 ($N_{\text{test}} = 533{,}379$):** AUROC $= 0.9963$, Hostile Rec $= 99.93\%$, RedTeam Rec $= 99.99\%$.

Decision-Theoretic Governance (Corollary 1):

- Reduces CMR by -18.45 pp (OTRF) and -37.32 pp (SALAD) and operational loss by 30.2% to 60.8%.
- Exact crossover $\kappa^* = 39.64$ (OTRF), 1.68 (SALAD); interior habituation optimum $\tau^*_{\text{LLR}} = 1.0$.